

TrES-5b

R-0922

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250710 · created 2026-07-29T09:09:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460866.7945 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.027
Transit depth (Rp/Rs)^2	3.2 +/- 0.98 [%]
Orbital Inclination (inc)	85.41 +/- 1.93
Airmass coefficient 1 (a1)	1.193 +/- 0.04
Airmass coefficient 2 (a2)	-0.15 +/- 0.072
Scatter in the residuals of the lightcurve fit is	3.38 %
Best Comparison Star	#2 - [532, 289]
Optimal Method	PSF photometry
Transit Duration (day)	0.0802 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.179 ± 0.027 vs 0.142 (deviation 1.37σ)
Duration fit vs published	0.0802 d vs 0.0758 d (deviation 0.47σ)
Mid-time O–C	1.4 min
Depth SNR	6.6
Residual scatter	3.38 %

Light curve

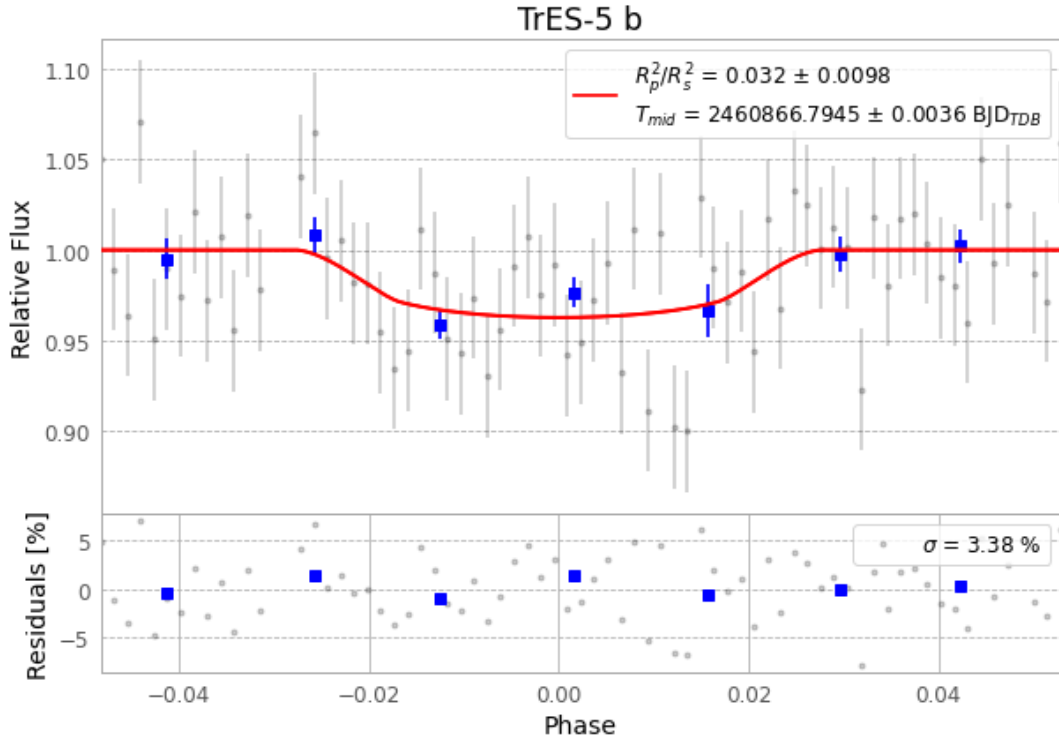


Figure 1 — FinalLightCurve_TrES-5 b_2025-07-09.png (SHA-256 0c7c755734dc8720...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 208], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6d59b90a697ce8c9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), TRES-5250710051816.FITS → TRES-5250710085415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250710.log (886fc11b8b6a...), FinalLightCurve_TrES-5 b_2025-07-09.png (0c7c755734dc...), FinalLightCurve_TrES-5 b_2025-07-09.csv (3ba84bf5ce70...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 09:09Z.